CHAPTER 1

DESIGN A PROXIMITY SERVICE/YELP

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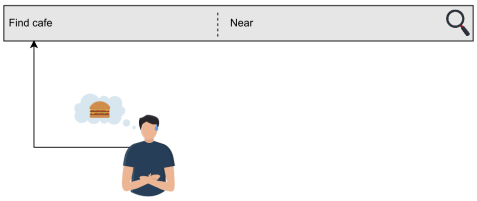
Take a closer look at the System Design of a proximity service like Yelp, addressing requirements like searching, scaling, and dynamic segments.

1 System Design: Yelp

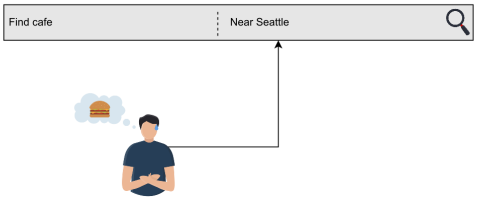
1.1 What is Yelp?

Yelp is a one-stop platform for consumers to discover, connect, and transact with local businesses. With it, users can join a waitlist, make a reservation, schedule an appointment, or purchase goods easily. Yelp also provides information, photos, and reviews about local businesses.
The user provides the name of a place or its GPS location, and the system finds places that are nearby. The user can also upload their opinions on this platform in the form of text, pictures, or ratings for a place they visited. Other location-based services include Foursquare and Google Nearby.
Services based on **proximity servers** are helpful in finding nearby attractions such as restaurants, theaters, or recreational sites. Designing such a system is challenging because we have to efficiently find all the possible places in a given radius with minimum latency. This means that we have to narrow down all the locations in the world, which could be in the billions, and only pinpoint the relevant ones.

(a) The user can search for a specific place near them



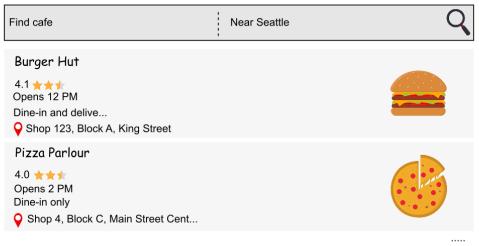
(b) A user enters "cafe" as their input



(a) A user wants to search for cafes around Seattle

(b) The proximity server will search for the places in a given radius, keeping user's location as centre point

(a) The proximity server will identify possible places that fit the user's input



(b) The proximity server will return the relevant places around Seattle to the user

1.2 How will we design Yelp?

Here is the breakdown of Yelp's design:

1. **Requirements**: In this lesson, we define the requirements and estimate the required servers, storage, and bandwidth of our system.
2. **Design**: In this lesson, we define the API design, the database schema, the components of our system, and the workflow of Yelp.
3. **Design considerations**: In this lesson, we dive deep into the design of the Yelp system.
4. **Quiz**: In this lesson, we take a quiz to test our knowledge of Yelp design.

Let's start our design by defining its requirements.

2 Requirements of Yelp's Design

2.1 Requirements

Let's identify the requirements of our system.

Functional requirements

The functional requirements of our systems are below:

- **User accounts**: Users will have accounts where they're able to perform different functionalities like log in, log out, add, delete, and update places' information.

  Note: There can be two types of users: business owners who can add their places on the platform and other users who can search, view, and give a rating to a place.

- **Search**: The users should be able to search for nearby places or places of interest based on their GPS location (longitude, latitude) and/or the name of a place.
- **Feedback**: The users should be able to add a review about a place. The review can consist of images, text, and a rating.

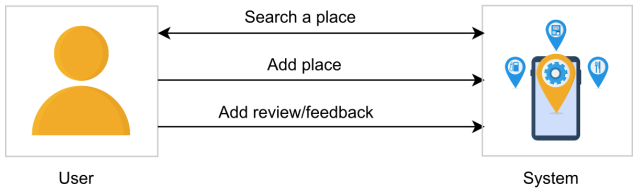


Figure 1.4: Functional requirements

Non-functional requirements

The non-functional requirements of our systems are:

- **High availability**: The system should be highly available to the users.

- **Scalability**: The system should be able to scale up and down, depending on the number of requests. The number of requests can vary depending on the time and number of days. For example, there are usually more searches made at lunchtime than at midnight. Similarly, during tourist season, our system will receive more requests as compared to in other months of the year.
- **Consistency**: The system should be consistent for the users. All the users should have a consistent view of the data regarding places, reviews, and images.
- **Performance**: Upon searching, the system should respond with suggestions with minimal latency.

2.2 Resource estimation

Let's assume that we have:

- A total of 178 million unique users.
- 60 million daily active users.
- 500 million places.

Number of servers estimation

Considering our assumption (from Back-of-the-envelope Calculations) of using daily active users as a proxy for the number of requests per second, we get 60 million requests per second. Then , we use the following formula to calculate the number of servers:

$$\text{Servers needed at peak load} = \frac{\text{Number of requests/second}}{\text{RPS of server}}$$

Using 64,000 as an estimated RPS a server can handle, the required servers at peak times are estimated as follows:

$$\text{Servers needed at peak load} = \frac{60 \text{ million}}{64,000} = 937.5 \approx 1\text{K servers}$$

Figure 1.5: The number of servers required for Yelp

Storage estimation

Let's calculate the storage we need for our data. Let's make the following assumptions:

- We have a total of 500 million places.
- For each place, we need 1,296 Bytes of storage.
- We have one photo attached to each place, so we have 500 million photos.
- For each photo, we need 280 Bytes of storage. Here , we consider the row size of the photo entity in the table, which contains a link to the actual photo in the blob store.
- At least 1 million reviews of different places are added daily.
- For each review, we need 537 Bytes of storage.

- We have a total of 178 million users.
- For each user, we need 264 Bytes of storage.

  Note: The Bytes used for each place, photo, review, and user are based on the database schema that we'll discuss in the next lesson.

The following calculator computes the total storage we need:

Figure 1.6: The total amount of storage required by Yelp

Note: For reviews, the estimate of 0.54 GB represents only a single day's reviews. For long-term storage over 10 years, we need to multiply this by the number of days (365×10), giving us approximately 2 TB, which is a more realistic storage requirement. This will also increase the total storage to approximately 2.8 TB.

Bandwidth estimation

To estimate the bandwidth requirements for Yelp, we categorize the bandwidth calculation of incoming and outgoing traffic.
For incoming traffic, let's assume the following:

- On average, five places are added every day.
- For each place, we take up 1,296 Bytes.
- A photo of size 3 MB is also attached with each place. This is the size of the photo that we save in the blob store.
- One million reviews of different places are added every day.
- Each review, takes up 537 Bytes.

We divide the total size of information per day by 86,400 to convert it into per second bandwidth.

For outgoing traffic, let's assume the following:

- A single search returns 20 places on average.
- Each place has a single photo attached to it that has an average size of 3 MB.
- Every returned entry contains the place and photo information.

Considering that there are 60 million active daily users, we come to the following estimations:

We need a total of approximately 51 Kbps of incoming traffic and approximately 331 Gbps of outgoing, assuming that the uploaded content is not compressed.

Total bandwidth requirements $= 51 \text{ Kbps} + 331 \text{ Gbps} \approx 331 \text{ Gbps}$.

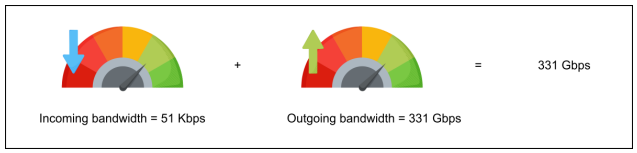


Figure 1.7: The total bandwidth required by Yelp

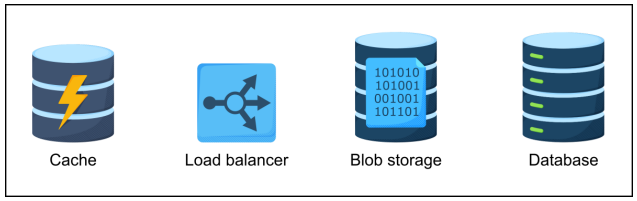


Figure 1.8: Building blocks in the high-level design of Yelp

2.3 Building blocks we will use

The design process of Yelp utilizes many building blocks that have already been discussed in the initial chapters of the course. We 'll consider the following concepts while designing Yelp:

- **Caching**: We'll use the cache to store information about popular places.
- **Load balancer**: We'll use the load balancer to manage the large number of requests.
- **Blob storage**: We'll store images in the blob storage.
- **Database**: We'll store information about places and users in the database.

We'll also rely on **Google Maps** to understand the feature of searching for places within a particular radius.

3 Design of Yelp

We identified the requirements and calculated the estimations for our Yelp system in the previous lesson. In this lesson, we discuss the API design, go through the storage schema, and then dive into the details of the system's building blocks and additional components.

3.1 API design

Let's discuss the API design for Yelp.

Search

We need to implement the `search` function. The API call for searching based on categories like "cafes" will be:

```
search(category, user_location, radius)
```

This process returns a JSON object that contains a list of all the possible items in the specified category that also fall within the specified radius. Each entry has a place name, address, category, rating, and thumbnail.
The API call for searching based on the name of a place like "Burger Hut" will be:

Parameter	Description
category	This is the type of search the user makes—for example, a search for restaurants, cinemas, cafes, and so on.
user_location	This contains the location of the user who's searching with Yelp.
radius	This is the specified radius where the user is trying to find the required category.

```
search(name_of_place, user_location, radius)
```

Parameter	Description
name_of_place	This contains the name of the place that the user wants to search for.

This process returns a JSON object that contains information of the specified place.

Add a place

The API call for adding a place is below:

```
add_place(name_of_place, description_of_place, category, latitude, longitude,
    photo}
```

Parameter	Description
name_of_place	This contains the name of the place, for example, "Burger Hut".
description_of_place	This contains a description of the place. For example, "Burger Hut sells the yummiest burgers".
category	This specifies the category of the place—for example, "cafe".
latitude	This tells us the latitude of the place.
longitude	This tells us the longitude of the place.
photo	This contains photos of the place. There can be a single or multiple photos.

This process returns a response saying that a place has been added, or an appropriate error if it fails to add a place.

Add a review

The API call for adding a review is below:

```
add_review(place_ID, user_ID, review_description, rating)
```

This process returns a response that a review has been added, or an appropriate error if it fails to add a review.

3.2 Storage schema

Let's define the storage schema for our system. A few of the tables we might need are "Place," "Photos," "Reviews," and "Users."
Let's define the columns of the "Place" table:

- **Place_ID**: We use the sequencer to generate an 8 Bytes (64 bits) unique ID for a place.

  Note: We generate IDs using the unique ID generator.

- **Name_of_Place**: This is a string that contains the name of the place. We use 256 Bytes for it.

Parameter	Description
`place_ID`	This contains the ID of the place whose review is added.
`user_ID`	This contains the ID of the user who adds the review.
`review_description`	This contains the review of the place—for example, "the food and ambiance were superb".
`rating`	This contains the rating of the place—for example, 4 out of 5.

- **Description_of_Place**: This holds a description of the place. We use 1,000 Bytes for it.
- **Category**: This specifies the type of place like restaurants, cinemas, bookshops, and so on (8 Bytes).
- **Latitude**: This stores the latitude of the location (8 Bytes).
- **Longitude**: This stores the longitude of the location (8 Bytes).
- **Rating**: This stores the rating of the place. It shows how many stars a place gets out of five. The rating is calculated based on the reviews it gets from the users.

The columns mentioned above are the most important ones in the table. We can add more columns like "menu," "address," "opening and closing hours," and so on. Therefore, keeping in mind the essential columns, the size of one row of our table will be:

```
Size = 8 + 256 + 1000 + 8 + 8 + 8 + 1 = 1289 bytes
```

Now let's define the "Photos" table:

- **Photo_ID**: We use the sequencer to generate a unique ID for a photo (8 Bytes or 64 bits).
- **Place_ID**: We use the foreign key (8 Bytes) from the "Place" table to identify which photo belongs to which place.
- **Photo_path**: We store the photos in blob storage and save the photo's path (256 Bytes) in this column.

```
Size = 8 + 8 + 8 + 256 = 280 bytes
```

We need another table called "Reviews" to store the reviews, ratings, and photos of a place.

- **Review_ID**: We use the sequencer to generate a unique ID of 8 Bytes (64 bits) for a review.
- **Place_ID**: The foreign key (8 Bytes) from the "Place" table to determine which place the rating belongs to.
- **User_ID**: The foreign key (8 Bytes) from the "Users" table to identify which review belongs to which user.
- **Review_description**: This holds a description of the review. We use 512 Bytes for it.
- **Rating**: This stores how many stars a place gets out of five (1 Byte).

```
Size = 8 + 8 + 8 + 512 + 1 = 537 bytes
```

We use the "Users" table to store user information.

- **User_ID**: We use the sequencer to generate a unique ID for a user (8 Bytes).
- **User_name**: This is a string that contains the user's name. We use 256 Bytes for it.

```
Size = 8 + 256 = 264 bytes
```

Note: The `INT` in the following schema contains an 8-Byte ID that we generate using the unique ID generator.

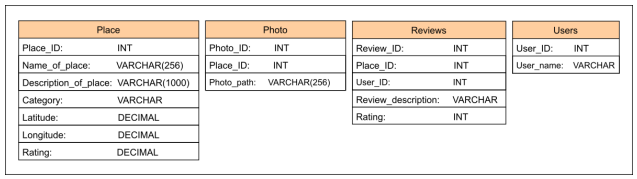


Figure 1.9: Storage schema

3.3 Design

Now we'll discuss the individual building blocks and components used in the design of Yelp and how they work together to complete various functional requirements.

Components

These are the components of our system:

- **Segments producer**: This component is responsible for communicating with the third-party world map data services (for example, Google Maps). It takes up that data and divides the world into smaller regions called segments. The segment producer helps us narrow down the number of places to be searched.
- **QuadTree servers**: These are a set of servers that have trees that contain the places in the segments. A QuadTree server finds a list of places based on the given radius and the user's provided location and returns that list to the user. This component mainly aids the search functionality.
- **Aggregators**: The QuadTrees accumulate all the places and send them to the aggregators. Then, the aggregators aggregate the results and return the search result to the user.
- **Read servers**: We use a set of read servers that we use to handle all the read requests. Since we have more read requests, it's efficient to separate these requests from the write requests. Each read server directs the search requests to the QuadTrees' servers and returns the results to the user.
- **Write server**: We use a set of write servers to handle all the write requests. Each write server handles the write requests of the user and updates the storage accordingly. Examples for write requests include adding a place, writing a comment, rating a place, and so on.
- **Storage**: We'll use two types of storage to fulfill our diverse needs.
  - **SQL database**: Our system will have different tables like "Users," "Place," "Reviews," "Photos," and others as described below. The data in these tables is inherently relational and structured. We need to perform queries like places a user visited, reviews they added, or view all the reviews of a specific place. It's easy to perform such queries in a SQL-based database. We also want all users to have a consistent view of the data, and SQL-based databases are better suited for such use cases. We'll use reliable and scalable databases, as is discussed in the Database building block.
  - **Key-value stores**: We'll need to fetch the places in a segment efficiently. For that, we store the list of places against a segment ID in a key-value store to minimize searching time. We also save the QuadTree information in the key-value store, by storing the QuadTree data against a unique ID.

- **Load balancer**: A load balancer distributes users' incoming requests to all the servers uniformly.

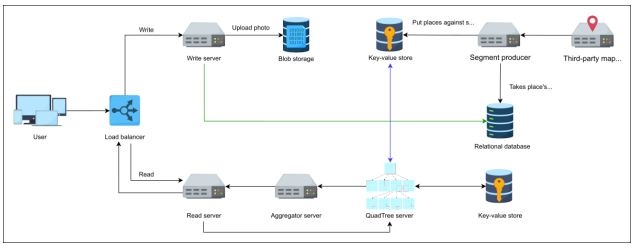


Figure 1.10: The detail design of Yelp

Workflow

The user puts in a search request. We find all the relevant places in the given radius, while considering the user's location (latitude, longitude).
We explain the detailed workflow of our system in terms of the required functionalities below:

- **Searching a place**: The load balancers route read requests to the read servers upon receiving them. The read servers direct them to the QuadTree servers to find all the places that fall within the given radius. The QuadTree servers then send the results to the aggregators to refine them and send them to the user.
- **Adding a place or feedback**: The load balancers route the write requests to the write servers upon receiving them. Depending on the provided content, meaning the place information or review, the write servers add an entry in the relational database and put all the related images in the blob storage.
- **Making segments**: The segment's producer splits the world map taken from the third-party map service into smaller segments. The places inside each segment are stored in a key-value store. Even though this is a one-time job, this process is repeated periodically for newer segments and places. Since the probability of new places being added is low, we update our segments every month.

We've discussed the design of Yelp, its API design, and the relevant storage schema. In the next lesson, we'll talk about the design considerations.

4 Design Considerations of Yelp

4.1 Introduction

We discussed the design, building blocks, components, and the entire workflow of our system in the previous lesson, which brought up a number of interesting questions that now need answering. For example, what approach do we use to find the places, or how do we rank places based on their popularity? This lesson addresses important design concerns like the ones we just mentioned. The table given below summarizes the goals of this lesson.

Summary of the Lesson

4.2 Searching

From Google Maps, we were able to connect segments and meet the scalability challenge to process a large graph efficiently. The graph of the world had numerous nodes and vertices, and traversing them was time-consuming. Therefore, we divided the whole world into smaller segments/subgraphs to process and query

Section/Sub-section	Purpose
Searching	This process divides the world into segments to optimize the search, so that all nearby sites in a given location and radius can be identified.
Storing Indexes	We index the places to improve query performance and estimate the storage required for all of the indexes.
Searching Using Segments	We search all the desired places by combining the segments.
Dynamic Segments	We solve the static segment limitations using dynamic segments.
Searching Using a QuadTree	We explore the searching functionality using a QuadTree.
Space Estimations for a QuadTree	We estimate the storage required for a QuadTree.
Data Partitioning	We look into the options we can use to partition data.
Ensuring Availability	We look into how we'll ensure the availability of the system.
Inserting a New Place	We look into how we'll insert the place in a QuadTree.
Ranking Popular Places	We rank the places of the system.
Evaluation	We evaluate how our system fulfills the non-functional requirements.

Table 1.1: Summary of the Lesson

them simultaneously. The segmentation helps us improve the scalability of the system.

Each segment will be of the size 5×5 miles and will contain a list of places that exist within it. We only search a few segments to locate destinations that are close by. We can use a given location and defined radius to find all the nearby segments and identify sites that are close.

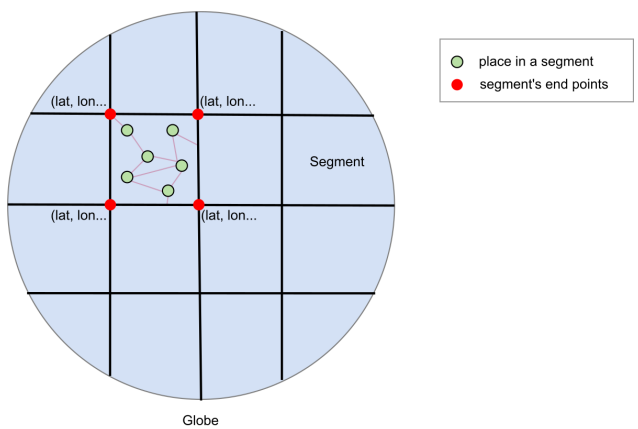


Figure 1.11: Partitioning the globe into small segments, where each segment has four coordinates (lat, long) and all segments hold the coordinates of different places

Quiz: Question 1: How will we find nearby places if we create a table for storing all places? **Answer:** We can make a table of places that have 'Place_ID' as a unique ID and store each place in it. The longitude and latitude columns help us specify the exact location. Indexing both of these columns can help us fetch data efficiently. We can specify each location as a pair of latitude ('M') and longitude ('N'). We can also search for a place within a given radius of 'R' by finding all the places between the latitude 'M-R' and 'M+R' and the longitude 'N-R' and 'N+R'. We can apply

Dijkstra's algorithm to find the distance between two points. **Question 2:** How efficient will our searching be when based on the table? **Answer:** We can have multiple lists of places within ('M-R', 'M+R') and ('N-R',

'N+R'). It'll be a challenge to handle concurrent requests. When queries that are used to fetch the places will be coming at a high rate for different segments, the response time and performance will be affected.

We can store all the places in a table and uniquely identify a segment by having a segment_ID. We can index each segment in the database. Now , we have limited the number of segments we need to search, so the query will be optimized and return results quickly.

Improve data fetching

We use a key-value store for quick access to places in the segments. The key is the segment_ID, while the `value` contains the list of places in that segment. Let 's estimate how much space we need to store the indexes.

> We can usually store a few MBs in the value of the key-value store. If we assume that each segment has 500 places then it will take up 500×1296 Bytes $= 0.648$MB, and we can easily store it as a value.

The total area of the earth is around 200 million square miles, and the land surface area is about 60 million square miles. If we consider our radius to be ten miles, then we'll have 6 million segments. An 8-Byte identifier will identify each segment.
Let's calculate how much memory we need.
Component type 'InstaCalc' not yet supported.

Search using segments

A user may select a radius for searching places that aren't present in a single segment. So , we need to combine multiple segments by connecting the segments to find locations within the specified radius—say, five miles.
First, we constrain the number of segments. This reduces the graph size and makes the searching process optimizable. Then , we identify all the relevant locations, compute the distance from the searching point, and show it to the user.

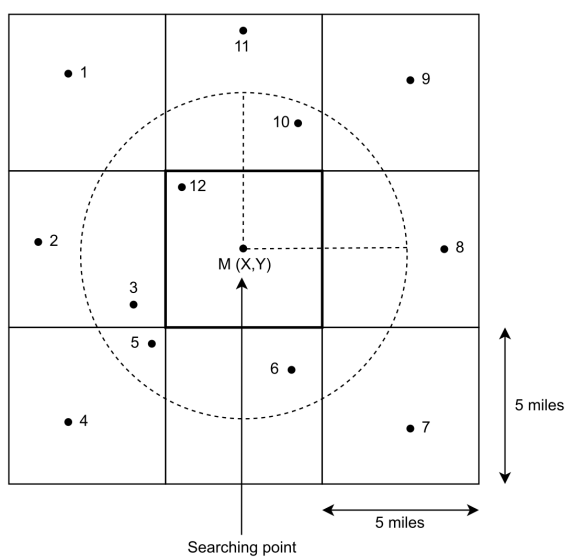


Figure 1.12: Searching for locations using segments

Quiz: Question 1: Can you identify a problem with the current approach? **Answer:** Our locations are not evenly distributed across segments, and this approach may still take a long time to run on segments with a lot of places. For example, in a city like New York, we can have lots of places even within a small radius because it is a densely-populated area. And for less populated areas, the same radius might not have enough places,

and we might need to expand our radius to find more.

4.3 Dynamic segments

We solve the problem of uneven distribution of places in a segment by dynamically sizing the segments. We do this by focusing on the number of places. We split a segment into four more segments if the number of places reaches a certain limit. We assume 500 places as our limit. While using this approach, we need to decide on the following questions:

1. How will we map the segments?

2. How will we connect to other segments?

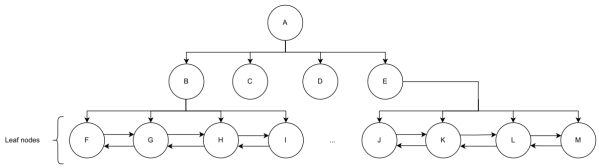
We use a **QuadTree**(A QuadTree is a tree data structure in which each internal node has exactly four children. QuadTrees are the two-dimensional analog of octrees and are most often used to partition a two-dimensional space by recursively subdividing it into four quadrants or regions. The data associated with a leaf cell varies by application, but the leaf cell represents a "unit of interesting spatial information". Source: Wikipedia) to manage our segments. Each node contains the information of a segment. If the number of places exceeds 500, then we split that segment into four more child nodes and divide the places between them. In this way, the leaf nodes will be those segments that can't be broken down any further. Each leaf node will have a list of places in it, too.

Search using a QuadTree

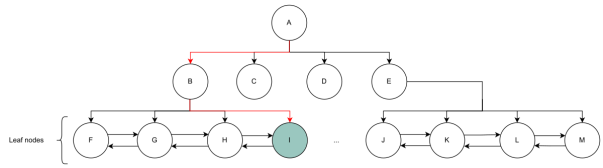
We start searching from the root node and continue to visit the nodes to find our desired segment. We check every node to see if it has more child nodes. If a node has no more children, then we stop our search because that node is the required one. We also connect each child node with its neighboring nodes with a doubly-linked list. All the child nodes of all the parent nodes are connected through the doubly-linked list. This list allows us to find the neighboring segments when we can move forward and backward as per our requirement. After identifying the segments, we have the required `PlaceID` values of the places and we can search our database to find more details on them.

> **Quiz: Question 1:** Is there an alternative approach to find the neighboring segments? **Answer:** We can use the pointers of parent nodes to find the neighboring segments. In each node, we can keep a pointer that points towards the parent node. Every parent node has pointers to its children nodes, so we can use those to find the adjacent leaf nodes. We can extend our search by going up through the parent pointers. In the following illustration,
>
> node 'F' can find its neighboring node, 'G', by first going to the parent node, 'B', and then to 'G'.
>
> ![](/api/collection/10370001/4941429335392256/page/5518783229198336/image/4551265476673536?page_type=collect

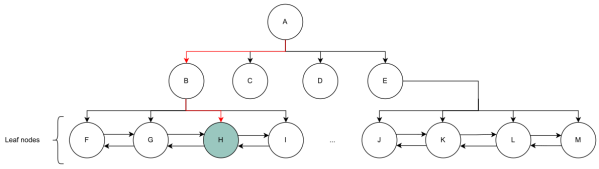
The following slides show how the process of searching for a place works. If a node has the places we need, we stop there. Otherwise, we explore more nodes until we reach our search radius. After finding the node, we query the database for information related to the places and return the desired ones.



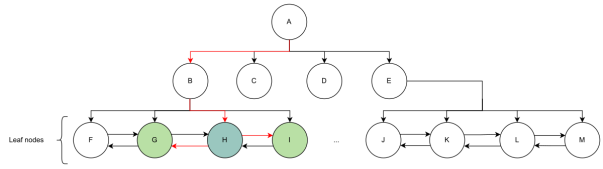
(a) All the leaf nodes in the QuadTree are linked, just like in a doubly-linked list



(b) We searched for a cafe and found the required number cafes—say, 20—in the same leaf node, so do not need to iterate any other nodes



(a) We searched for a cafe but only found, say, six cafes in a leaf node



(b) The leaf node will explore its neighboring nodes to find more cafes

Storage space estimation for QuadTrees

Let's calculate the storage we need for keeping QuadTrees:

- We store the `PlaceID`, `Latitude`, and `Longitude` for each place in the node. Each of these values is of the size 8 Bytes. For 500 million places, we need the following amount of storage for all places:
  - $(8 + 8 + 8) \times 10^6 \times 500 = 12\text{GB}$
- We'll have 500 places in a single node. So , for 500 million places, we'll need $\frac{500M}{500} = 1\text{M}$ leaf nodes.
- The QuadTree with 1 million leaf nodes has approximately 1/3rd of the leaf nodes of the internal nodes(Source: https://tildesites.bowdoin.edu/~ltoma/teaching/cs3225-GIS/fall16/Lectures/gis_qdt1.pdf) (all the nodes up the leaf level), and each internal node has four pointers to its child nodes. If we'll assume 8 Bytes for each pointer, then we'll need $1\text{M} \times 1/3 \times 4 \times 8 = 10.67\text{MB}$ of space for the internal nodes.
- We can get the total space needed to store all the internal nodes by adding 12GB and 10.67MB, that is, 12.01GB approximately.

We can easily store a QuadTree on a server.
Let's try the following calculator to calculate the storage needed for a QuadTree:
Component type 'InstaCalc' not yet supported.

Data partitioning

Keeping 20% growth per year in mind, the number of places will increase. We can partition data on the following basis:

1. **Regions**: We can split our places into regions on the basis of zip codes. This way, all the places that belong to a specific region are stored on a single node. We store information on the region along with the place, so that we can query on the basis of regions too. We can use the user's region to find the places in that specific region. This data partitioning comes with a few challenges. For example, if a region becomes popular during tourist season, it can affect the performance of our system. We might have numerous queries on the server that might result in slow responsiveness to user queries.
2. `PlaceID`: We can partition data on the basis of `PlaceID` instead of the region to avoid the query overload in popular seasons or rush hours. We can use a key-value store to store the places. In this case, the key is the `PlaceID` and the value contains the server in which that place is stored. This will make the process of fetching places more efficient.

So, we'll opt for partitioning on the basis of places. Moreover, we'll also use caches for popular places. The cache will have information about that particular place.

Ensure availability

Consider a scenario where multiple people in the same radius place a search request. If we have a single QuadTree, it'll affect the availability of the users. So, we can't rely on a single QuadTree. To cater to this problem, we replicate our QuadTrees on multiple servers to ensure availability. This allows us to distribute the read traffic and decrease the response time. Quad Trees are built on a server, so we can use the server ID as a key to identify the server on which the QuadTree is present. The value is the list of places that the QuadTree holds. The key-value store eases the rebuilding of the QuadTree in case we lose it.

> **Quiz: Question 1:** How will the leader-follower approach help us in replication? **Answer:** We'll have a single leader and two followers. The leader will have the
>
> QuadTree and it'll handle all the write requests. It'll update the followers about any change made to the QuadTrees synchronously, so there'll be a chance of delay. The followers will handle all the read requests. In case a follower dies, we'll choose another follower and replicate the data to it. If the leader is down, we'll choose any of the followers to step in as the leader. This way, we'll be able to replicate our QuadTrees onto three servers.

Insert a new place

We insert a new place into the database as well as in our QuadTree. We find the segment of the new place if the QuadTree is distributed on different servers, and then we add it to that segment. We split the segment if required and update the QuadTree accordingly.

Rank popular places

We need a service, a **rating calculator**, which calculates the overall rating of a service. We can store the rating of a place in the database and also in the QuadTree, along with the ID, latitude, and longitude of the place. The QuadTree returns the top 50 or 100 popular places within the given radius. The **aggregator service** determines the actual top places and returns them to the user.

> **Note:** We don't expect the rating to be updated within hours, as such frequent changes in QuadTrees or databases can be expensive. We update them once a day, when the load is minimal.

4.4 Finalized design

We added a few new components to our design. We introduced caches to store popular places. This allows us to fetch the places much faster. Moreover , we also added a rating calculator that sorts the places based on their ratings. This will enhance user experience, since the places with a good rating will be displayed first. The updated design of our system is shown below:

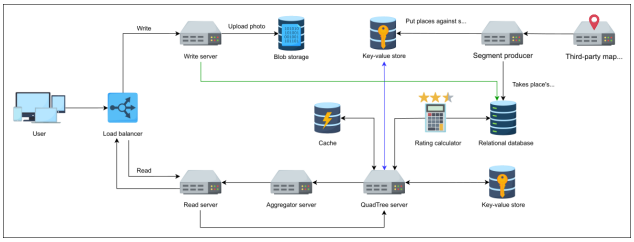


Figure 1.15: The updated detail design of Yelp

4.5 Evaluation

Let's see how our system design fulfills the non-functional requirements we identified in the initial lessons.

Requirements compliance

- **Availability**: We partitioned the data into smaller segments instead of having to deal with a huge dataset consisting of all the places on the world map. This made our system highly available. We also replicated the QuadTrees data using key-value stores to ensure availability.
- **Scalability**: We split the whole world into smaller dynamic segments. This allows us to search for a place within a specific radius and shorten our search area. We then used QuadTrees in which each child node holds a single segment. Upon adding or removing a place, we can restructure our QuadTrees. So, we were able to make our system scalable.
- **Performance**: We reduced the latency by using caches. We cached all the famous and popular places, so request time was minimized.
- **Consistency**: The users have a consistent view of the data regarding places, reviews, and photos because we used reliable and fault-tolerant databases like key-value stores and relational databases.

In the previous lesson, we chose a **SQL database** and a **key-value store** to design Yelp. How does replacing SQL with NoSQL impact the design? Discuss its impact on nonfunctional requirements like scalability and consistency. Also, discuss schema flexibility and the impact on the performance of complex queries.

Question:

How does replacing SQL with NoSQL impact the design?

Reference Answer:

SQL database: Yelp system will have different tables like "Users," "Place," "Reviews," "Photos," and others as described below. The data in these tables is inherently relational and structured. We need to perform queries like places a user visited, reviews they added, or view all the reviews of a specific place. It 's easy to perform such queries in a SQL-based database. We also want all users to have a consistent view of the data, and SQL-based databases are better suited for such use cases. We 'll use reliable and scalable databases, as is discussed in the Database building block.

Key-value stores: We'll need to fetch the places in a segment efficiently. For that, we store the list of places against a segment ID in a key-value store to minimize searching time. We also save the QuadTree information in the key-value store, by storing the QuadTree data against a unique ID.

NoSQL databases excel in scalability, allowing for horizontal scaling to handle large volumes of data and high traffic loads. However, they often prioritize availability and partition tolerance over strict consistency, leading to eventual consistency models that may result in temporary data inconsistencies. No SQL databases offer schema flexibility, enabling dynamic data models without predefined schemas, but they may struggle with complex queries compared to SQL databases, which excel in supporting complex data analysis and reporting. Ultimately , the choice between SQL and NoSQL should consider the specific requirements of the application and the trade-offs between scalability, consistency, schema flexibility, and query performance.

4.6 Summary

The proximity-based servers allow the user to search for a specific place or places nearby. The map data of the world is huge and dividing it into segments and finding the specific segment was a challenge in itself. So , we used QuadTrees to optimize our search and provided the user with a list of places with minimum latency.

5 Quiz on Yelp's Design

Quiz

Question 1: (Select all that apply.) Which requirement will not be fulfilled if we don't constrain the number of segments to reduce the graph size?

(Select all that apply.)

- **[CORRECT]** Availability
  Explanation: The query will return the results slower affecting the user response time.
- Consistency
- **[CORRECT]** Scalability
  Explanation: It won't be easy to modify a humongous graph.
- Reliability

Question 2: (True or False) The type of partitioning (region- or `PlaceID`-based) affects the structure of QuadTrees.

- **[CORRECT]** True
  Explanation: Yes, it does as we have no assurance that each segment will have the same number of places in it. Though this won't be an issue as we will be finding our neighboring segments within the defined radius.
- False

Question 3: In which scenario is there a greater chance of delay in response time if we use the leader-follower approach to help us in replication?

- When the leader updates the followers asynchronously
  Explanation: The leader will not wait until all followers have updated the data, it'll update the data in its storage and then send the update to followers.
- **[CORRECT]** When the leader updates the followers synchronously
  Explanation: The leader waits until all followers have updated the data.
- When the leader saves the updates of a day and sends the changes to followers once a day
  Explanation: The leader will not ask the followers to update the data at once so there won't be a delay.

Question 4: (Select all that apply.) What will happen if the leader and followers both die simultaneously?
(Select all that apply.)

- **[CORRECT]** We would have to rebuild the QuadTrees.
  Explanation: We would have to rebuild QuadTrees from the key-value stores where we kept them.
- **[CORRECT]** The performance will be affected.
  Explanation: We would not be able to serve any user query which will affect the performance.
- **[CORRECT]** The availability will be affected.
  Explanation: Both leader and followers will be down so they will be unable to process the requests.